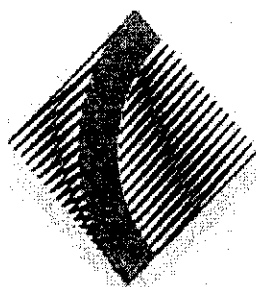


Name: _____

Class: 12MTX _____

Teacher: _____

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2004

YEAR 12

AP4 EXAMINATION

MATHEMATICS EXTENSION 1

*Time allowed - 2 HOURS
(Plus 5 minutes' reading time)*

DIRECTIONS TO CANDIDATES:

- Attempt all questions.
- All questions are of equal value.
- Each question is to be commenced on a new page clearly marked Question 1, Question 2, etc on the top of the page. **
- Each question is to be returned in a separate bundle.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.

****Each page must show your name and your class. ****

Question 1 (12 marks)**Marks**

- (a) The point $(-6t, 9t^2)$, where t is a variable, lies on a curve. Find the Cartesian equation of the curve. **2**
- (b) Two of the roots of the equation $x^3 - 13kx^2 + 13kx - 1 = 0$ are k and $\frac{1}{k}$ where $k \neq 0$.
- (i) Find the third root. **1**
- (ii) Find the value(s) of k . **2**
- (c) Solve: $\frac{x}{x-2} \geq 4, \quad x \neq 2$ **3**
- (d) For the expansion of the expression $\left(x - \frac{3}{x}\right)^5$, find the coefficient of x . **2**
- (e) Find $\frac{d}{dx}(2x^3 \tan^2 x)$. **2**

Question 2 (12 marks)

(Please start on a new page)

Marks

(a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 4x}{5x}$. **1**

(b) Find $\int \sin^2 3x \, dx$. **2**

(c) (i) Prove the identity **1**

$$\frac{\sin 2\theta}{2\sin \theta} - \cos \theta \cos 2\theta = 2\cos \theta \sin^2 \theta$$

(ii) Hence solve the equation **2**

$$\frac{\sin 2\theta}{2\sin \theta} - \cos \theta \cos 2\theta = \cos \theta \text{ for } 0 \leq \theta \leq 2\pi.$$

(d) Use mathematical induction to prove that, for all positive integers n , **4**

$$\sum_{r=1}^n \frac{r^2}{(2r-1)(2r+1)} = \frac{n(n+1)}{2(2n+1)}$$

(e) Differentiate $\log_7 x^2$ with respect to x . **2**

Question 3 (12 marks)

(Please start on a new page)

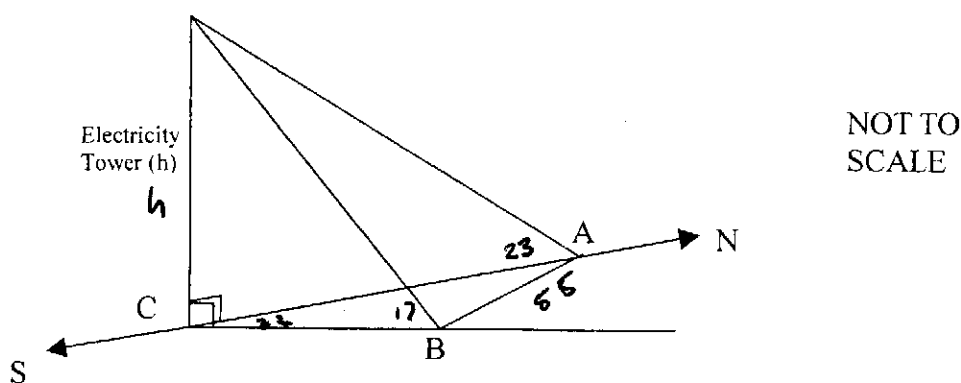
Marks

(a) Use the substitution $u = x - 2$ to evaluate $\int_3^4 \frac{x}{\sqrt{x-2}} dx$. 3

- (b) A person walks on level ground, in a northerly direction, away from an electricity tower. When they arrive at a point A, the angle of elevation to the top of the tower is 23° . Another person walks on level ground on a bearing of $032^\circ T$ from the same tower, until they reach point B, and measures the angle of elevation to the top of the tower as 17° .

It is known that the points A and B are 55 metres apart. Let h be the height of the tower and assume that the tower, with base C, is perpendicular to the ground.

- (i) Copy the diagram below onto your page, clearly marking all information. 1



- (ii) Find the distances AC and BC , leaving your answers in terms of h . 2
- (iii) Hence, or otherwise, find the height h of the tower, correct to the nearest metre. 3
- (c) Consider the geometric series $\sin 2x + \sin 2x \cos 2x + \sin 2x \cos^2 2x + \dots$ 3
for $0 < x < \frac{\pi}{2}$. Show that the limiting sum S of the series exists and that $S = \cot x$.

Question 4 (12 marks) *(Please start on a new page)* **Marks**

(a) (i) Write equation for the asymptotes of the curve $y = \ln(x - 2)$. **2**

(ii) The inner surface of a bowl is of the shape formed by rotating About the y axis, the curve $y = \ln(x - 2)$ between $y = 0$ and $y = 2$. The bowl is placed with its axis vertical and water is poured in. Show that the volume of water in the bowl when it is filled to a depth h , where $h < 2$, is given by **2**

$$\pi(4h - 4\frac{1}{2} + 4e^h + \frac{1}{2}e^{2h}) \text{ unit}^3.$$

(iii) If the bowl is filled at the rate of $60 \text{ unit}^3 / \text{s}$, find the rate at which the water level is rising when the depth of water is 1.25 units . Give your answer correct to 2 decimal places. **2**

(b) Corn cobs are cooked by immersing them in boiling water. On being removed, a corn cob cools in the air according to the equation $\frac{dT}{dt} = -k(T - T_0)$ where t is time in minutes, T is temperature in $^{\circ}\text{C}$ and T_0 is the temperature of the air, while k is a positive constant.

(i) Verify that $T = T_0 + Ae^{-kt}$ is a solution of the above equation where A is a constant. **2**

(ii) If the temperature of the boiling water is 100°C and that of the air is a constant 25°C , find the values of A and k if a corn cob cools to 70°C in 3 minutes. **2**

(iii) How long should a person wait to enjoy the food at a temperature of 50°C ? **2**

Question 5 (12 marks)

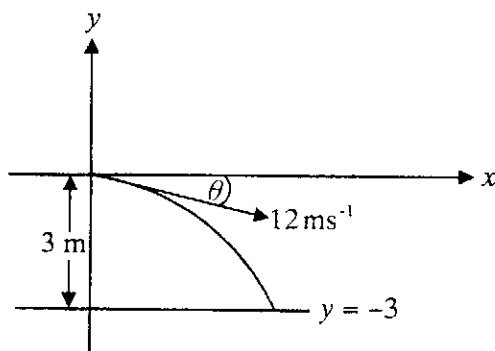
(Please start on a new page)

Marks

- (a) A particle moves with a simple harmonic motion. It starts from rest at a point 6 cm from the centre of motion O. The particle has a speed of 10 cm/s, when it passes through O.

- (i) Find the amplitude of the motion. 1
- (i) Find the period of motion. 2
- (ii) Find the acceleration after 3 seconds, correct to 2 significant figures. 3

(b)



A child in a tree throws a ball and gives it a velocity of 12 ms^{-1} at an angle of θ below the vertical. The height of the ball above the ground when it leaves the child's hand is 3 m. Take the origin to be the point where the ball leaves the child's hand. The position of the ball at time t seconds after the ball has been thrown and before it hits the ground is given by (x, y) . The equations of motion for the ball are

$$\ddot{x} = 0 \text{ and } \ddot{y} = -10.$$

- (i) Show that 3

$$x = 12t \cos \theta$$

$$\text{and } y = -12t \sin \theta - 5t^2$$

- (ii) Given that $\tan \theta = \frac{5}{12}$, find the horizontal distance 3
between the point where the ball leaves the child's hand
and the point where the ball hits the ground.

Question 6 (12 marks)

(Please start on a new page)

Marks

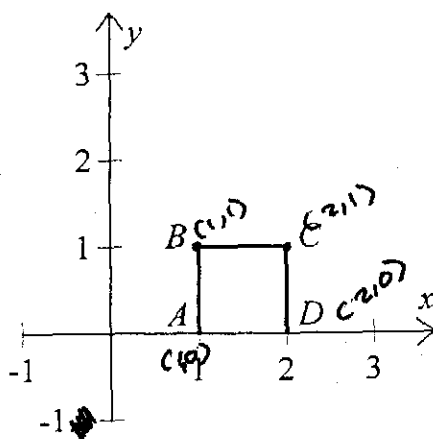
- (a) The point $P(2ap, ap^2)$ lies on the parabola $x^2 = 4ay$ whose focus is at S . The tangent at P meets the Y -axis at Q .

(i) Find the coordinates of Q . 1

(ii) Show that $\angle SPQ = \angle SQP$ 2

- (b) The acceleration of a particle is given by $a = -e^{-x}$. Initially $v = \sqrt{2}$ and $x = 0$. Find the velocity as a function of x . 2

- (c) The diagram shows a unit square $ABCD$, where $A(1, 0)$, $B(1, 1)$, $C(2, 1)$ and $D(2, 0)$.



Copy the diagram onto your answer sheet.

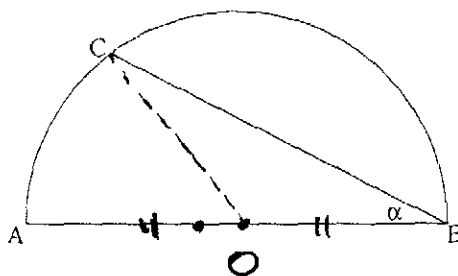
- (i) A line l , passing through the origin with gradient m , cuts the sides AB and CD at P and Q respectively. Comment on the possible values of m . 1
- (ii) For what value(s) of m does the line l divide the area of the square in the ratio 2:1. 3
- (iii) Another line K passes through the origin with gradient n , and cuts the square through sides AB and BC at S and T respectively. Show that it is not possible for k to divide the area of the square in the ratio 2:1. 3

Question 7 (12 marks)

(Please start on a new page)

Marks

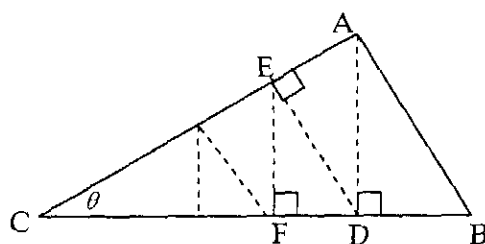
- (a) AB is the diameter of a semi-circle of unit radius with centre O and BC is a chord which makes an angle α with AB . The area of the semi-circle is bisected by this chord.



- (i) Write an expression for $\angle BOC$ in terms of α . 1
- (ii) Show that the area of the segment is $\frac{1}{2}(\pi - 2\alpha - \sin 2\alpha)$ 1
- (iii) Hence show that $2\sin 2\alpha + 4\alpha - \pi = 0$ 2
- (iv) Prove that a root of this equation lies between $\alpha = 0.4$, and $\alpha = 0.5$. 2
- (v) By using the 'halving the interval' method, determine whether the root lies closer to 0.4 or 0.5. 1

- (b) In the triangle ABC , $\angle ACB = \theta$, where $0 < \theta < \frac{\pi}{2}$ and AC is of length d .

A fly starts at A , flies directly to the line CB , i.e. to the point D . It then flies directly to the line CA , i.e. to the point E . It then flies directly to the line CB and so on until it ultimately reaches C .



- (i) Prove that $\angle ADE = \theta$. 1
- (ii) Show that the distance travelled by the fly when it reaches the point E is $d \sin \theta (1 + \cos \theta)$. 2
- (iii) Show that the total distance travelled is given by 2

$$s = \frac{d \sin \theta}{1 - \cos \theta}$$

END OF TEST

EXTENSION 1 AP1 2004

AMENDMENT

Question 4 (12 marks)	<i>(Please start on a new page)</i>	Marks
(a) (i)	Sketch the curve $y = \ln(x - 2)$, showing its asymptote.	2

(a) $x = -bt$

$t = -\frac{x}{b} \rightarrow \textcircled{1}$

$y = 9t^2$

$y = 9\left(-\frac{x}{b}\right)^2$

$y = \frac{x^2}{4} \rightarrow \textcircled{1}$

(b) Let the third root be α .

$k \times \frac{1}{k} \times \alpha = 1$

$\alpha = 1 \rightarrow \textcircled{1}$

(c) $k + \frac{1}{k} + 1 = 13k$

$k^2 + 1 + k = 13k^2$

$12k^2 - k - 1 = 0 \rightarrow \textcircled{1}$

$(4k+1)(3k-1) = 0$

$k = -\frac{1}{4} \rightarrow \textcircled{\frac{1}{2}}$

$k = \frac{1}{3} \rightarrow \textcircled{\frac{1}{2}}$

(c) $\frac{x^{x(x-2)^2}}{x-2} \geq 4(x-2)^2$

$x^2 - 2x \geq 4x^2 - 16x + 16$

$3x^2 - 14x + 16 \leq 0 \rightarrow \textcircled{1}$

$(x-2)(3x-8) \leq 0$

$2 \leq x \leq 2\frac{2}{3} \rightarrow \textcircled{1}$

$2 < x \leq 2\frac{2}{3} \rightarrow \textcircled{1}$

(d)

$x^5 - 5x^4 \cdot \frac{3}{x} + 10x^3 \left(\frac{3}{x}\right)^2 -$

$10x^2 \left(\frac{3}{x}\right)^3 + 5x \left(\frac{3}{x}\right)^4 - \left(\frac{3}{x}\right)^5$

$\rightarrow \textcircled{1}$

$= \dots + 10x^3 \times \frac{9}{x^2} - \dots$

$= \dots + 90x - \dots$

$\therefore$ Coefficient of x is 90 $\rightarrow \textcircled{1}$

(e) $6x^2 \tan^2 x + \underbrace{2x^3 (2 + \tan x) \sec^2 x}_{\rightarrow \textcircled{1}}$

$= 6x^2 \tan^2 x + 4x^3 \tan x \sec^2 x$

QUESTION 2

$$\begin{aligned}
 (a) \quad \lim_{x \rightarrow 0} \frac{\sin 4x}{4x} \times \frac{4}{5} \\
 = 1 \times \frac{4}{5} \\
 = \frac{4}{5} \rightarrow \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad \cos 2x &= 1 - 2\sin^2 x \\
 \cos 6x &= 1 - 2\sin^2 3x \\
 \sin^2 3x &= \frac{1}{2} - \frac{1}{2}\cos 6x \rightarrow \textcircled{1} \\
 \int \sin^2 3x \, dx &= \int \left(\frac{1}{2} - \frac{1}{2}\cos 6x \right) dx \\
 &= \frac{1}{2}x - \frac{1}{12}\sin 6x + C \rightarrow \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 (c) \textcircled{1} \textcircled{2} \text{ LHS} &= \frac{2\sin\theta \cos\theta}{2\sin\theta} \\
 &\quad - \cos\theta(1 - 2\sin^2\theta) \\
 &= \cos\theta - \cos\theta + 2\cos\theta\sin^2\theta \\
 &= 2\cos\theta\sin^2\theta \\
 &= \text{RHS} \rightarrow \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 (m) \quad 2\cos\theta \sin^2\theta &= \cos\theta \\
 \cos\theta(2\sin^2\theta - 1) &= 0 \\
 \cos\theta &= 0 \\
 \theta &= \frac{\pi}{2}, \frac{3\pi}{2} \rightarrow \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 2\sin^2\theta - 1 &= 0 \\
 \sin^2\theta &= \frac{1}{2} \\
 \sin\theta &= \pm \frac{1}{\sqrt{2}} \\
 \theta &= \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4} \rightarrow \textcircled{1}
 \end{aligned}$$

$$(d) \text{ Prove true for } n=1$$

$$\text{LHS} = \frac{1}{1 \times 3} = \frac{1}{3}$$

$$\text{RHS} = \frac{1 \times 2}{2 \times 3} = \frac{1}{3}$$

$$\text{LHS} = \text{RHS}$$

$$\therefore \text{ true for } n=1 \rightarrow \textcircled{1}$$

Assume true for $n=k$

$$\text{i.e. } \sum_{r=1}^k \frac{r^2}{(2r-1)(2r+1)} = \frac{k(k+1)}{2(2k+1)}$$

Hence prove true for $n=k+1$

$$\text{i.e. } \frac{k(k+1)}{2(2k+1)} + \frac{(k+1)^2}{(2k+1)(2k+3)} = \frac{(k+1)(k+2)}{2(2k+3)} \rightarrow \textcircled{1}$$

$$\begin{aligned}
 \text{LHS} &= \frac{k(k+1)(2k+3) + 2(k+1)^2}{2(2k+1)(2k+3)} \\
 &= \frac{(k+1)[k(2k+3) + 2(k+1)]}{2(2k+1)(2k+3)} \\
 &= \frac{(k+1)[2k^2 + 3k + 2k + 2]}{2(2k+1)(2k+3)} \\
 &= \frac{(k+1)[2k^2 + 4k + 2]}{2(2k+1)(2k+3)} \rightarrow \textcircled{1} \\
 &= \frac{(k+1)[2k(k+2) + 2(k+2)]}{2(2k+1)(2k+3)} \\
 &= \frac{(k+1)(k+2)(2k+1)}{2(2k+1)(2k+3)} \\
 &= \frac{(k+1)(k+2)}{2(2k+3)}
 \end{aligned}$$

$$= \text{RHS}$$

$\therefore$ if true for $n=k$ then true for $n=k+1$

QUESTION 2 (CONT.)

$\therefore$ if true for $n=1$ then true
for $n=2$, etc

$\therefore$ true for all integers $n \geq 1$
 $\rightarrow \textcircled{1}$

$$\begin{aligned} \text{(e)} \quad y &= \log_7 x^2 \\ &= 2 \log_7 x \\ &= \frac{2 \log_e x}{\log_e 7} \rightarrow \textcircled{1} \end{aligned}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{2}{\log_e 7} \cdot \frac{1}{x} \\ &= \frac{2}{x \log_e 7} \rightarrow \textcircled{1} \end{aligned}$$

QUESTION 3

(a) $u = x - 2$
 $du = dx$

when $x = 3$, $u = 1$
 $x = 4$, $u = 2$

$$\therefore \int_3^4 \frac{x}{\sqrt{x-2}} dx$$

$$= \int_1^2 \frac{u+2}{\sqrt{u}} du \rightarrow \textcircled{1}$$

$$= \int_1^2 (u^{1/2} + 2u^{-1/2}) du$$

$$= \left[\frac{2}{3} u^{3/2} + 4u^{1/2} \right]_1^2 \rightarrow \textcircled{1}$$

$$= \frac{4\sqrt{2}}{3} + 4\sqrt{2} - \left(\frac{2}{3} + 4 \right)$$

$$= \frac{16\sqrt{2}}{3} - 4\frac{2}{3}$$

$$= \frac{16\sqrt{2} - 14}{3} \rightarrow \textcircled{1}$$

OR

$$= \left[\frac{2}{3} (x-2)^{3/2} + 4(x-2)^{1/2} \right]_3^4$$

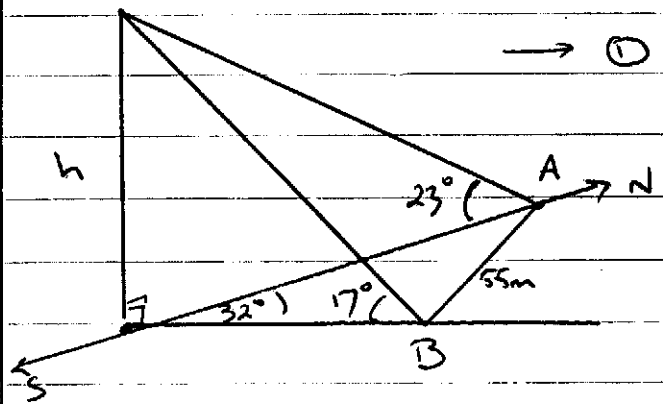
$$= \frac{4\sqrt{2}}{3} + 4\sqrt{2} - \left(\frac{2}{3} + 4 \right)$$

$$= \frac{16\sqrt{2}}{3} - 4\frac{2}{3}$$

$$= \frac{16\sqrt{2} - 14}{3} \rightarrow \textcircled{1}$$

Penalise $\textcircled{1}$ if boundaries do not match up with variable.

(b) (i)



(ii) $\tan 23^\circ = \frac{h}{AC}$

$$AC = \frac{h}{\tan 23^\circ} \rightarrow \textcircled{1}$$

$$\tan 17^\circ = \frac{h}{BC}$$

$$BC = \frac{h}{\tan 17^\circ} \rightarrow \textcircled{1}$$

(iii) Using Cosine Rule:

$$55^2 = \left(\frac{h}{\tan 23^\circ} \right)^2 + \left(\frac{h}{\tan 17^\circ} \right)^2 -$$

$$2 \left(\frac{h}{\tan 23^\circ} \right) \left(\frac{h}{\tan 17^\circ} \right) \cos 32^\circ \rightarrow \textcircled{1}$$

$$3025 = h^2 \left[\frac{1}{\tan^2 23^\circ} + \frac{1}{\tan^2 17^\circ} - \frac{2 \cos 32^\circ}{\tan 23^\circ \tan 17^\circ} \right]$$

$$h^2 = \frac{3025}{\left[\frac{1}{\tan^2 23^\circ} + \frac{1}{\tan^2 17^\circ} - \frac{2 \cos 32^\circ}{\tan 23^\circ \tan 17^\circ} \right]} \rightarrow \textcircled{1}$$

$$h^2 = 951.5569 \dots$$

$$h = 31 \text{ m (nm.)} \rightarrow \textcircled{1}$$

(-1 for incorrect accuracy)

QUESTION 3 (cont.).

$$(c) \quad r = \cos 2x \rightarrow \textcircled{1}$$

$$0 < |\cos 2x| < 1$$

$$\text{for } 0 < x < \frac{\pi}{2}$$

$\therefore$ a limiting sum exists $\rightarrow \textcircled{1}$

$$S = \frac{\sin 2x}{1 - \cos 2x}$$

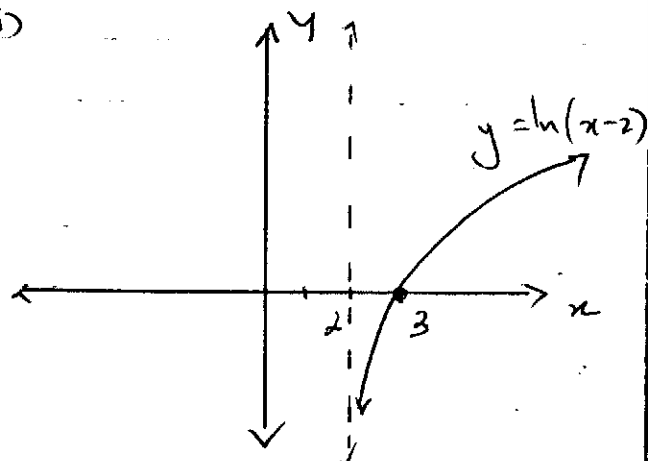
$$= \frac{2 \sin x \cos x}{2 \sin^2 x}$$

$$= \frac{\cos x}{\sin x}$$

$$S = \cot x \rightarrow \textcircled{1}$$

Question 4

a) (i)



① for asymptote

② for x-intercept

(ii) $x = e^y + 2$

$$V = \pi \int_0^h (e^y + 2)^2 dy \rightarrow \textcircled{1}$$

$$= \pi \int_0^h (e^{2y} + 4e^y + 4) dy$$

$$= \pi \left[\frac{e^{2y}}{2} + 4e^y + 4y \right]_0^h$$

$$= \pi \left[\frac{1}{2}e^{2h} + 4e^h + 4h - \frac{1}{2} + 4 \right]$$

$$= \pi \left[4h - 4\frac{1}{2} + 4e^h + \frac{1}{2}e^{2h} \right] \rightarrow \textcircled{2}$$

(iii) $\frac{dv}{dt} = 60$

$$\frac{dh}{dt} = \frac{dh}{dv} \cdot \frac{dv}{dt}$$

$$\frac{dV}{dh} = \pi (4 + 4e^h + e^{2h})$$

$$\therefore \frac{dh}{dt} = \frac{1}{\pi (4 + 4e^h + e^{2h})} \rightarrow \textcircled{1}$$

$$\begin{aligned} \frac{dh}{dt} &= \frac{1}{\pi (4 + 4e^{1.25} + e^{2.5})} \cdot 60 \\ &= 0.633581416 \dots \\ &= 0.63 \text{ units/s} \rightarrow \textcircled{1} \end{aligned}$$

b) (i) LHS = $\frac{dT}{dt} = -Ake^{-kt} \rightarrow \textcircled{1}$

$$\begin{aligned} \text{RHS} &= -k(T - T_0) \\ &= -k(T_0 + Ae^{-kt} - T_0) \\ &= -Ake^{-kt} \\ &= \text{LHS} \rightarrow \textcircled{1} \end{aligned}$$

(ii) $100 = 25 + Ae^0$
 $\therefore A = 75 \rightarrow \textcircled{1}$

$$\begin{aligned} 70 &= 25 + 75e^{-3k} \\ e^{-3k} &= \frac{3}{5} \end{aligned}$$

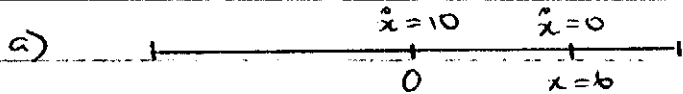
$$\begin{aligned} k &= \frac{\log \frac{3}{5}}{-3} \\ &= 0.170275207 \dots \rightarrow \textcircled{1} \\ &\doteq 0.1703 \text{ (4dp)} \end{aligned}$$

(iii) $50 = 25 + 75e^{-0.1703t}$
 $e^{-0.1703t} = \frac{1}{3} \rightarrow \textcircled{1}$

$$\begin{aligned} t &= \frac{\log \frac{1}{3}}{-0.1703} \end{aligned}$$

$$\textcircled{1} \leftarrow \begin{cases} \doteq 6.45104 \dots \\ \doteq 6.45 \text{ mins, (2dp)} \end{cases}$$

QUESTION 5.



$$\begin{aligned} \text{(i)} \quad v^2 &= n^2 (a^2 - x^2) \\ 0 &= n^2 (a^2 - 36) \\ a^2 &= 36 \\ a &= 6 \rightarrow \textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad 100 &= n^2 (36 - 0) \\ n &= \frac{5}{3} \rightarrow \textcircled{1} \end{aligned}$$

$$\begin{aligned} T &= \frac{2\pi}{n} = \frac{2\pi \times 3}{5} \\ &= \frac{6\pi}{5} \rightarrow \textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad x &= a \cos(ut + \alpha) \\ x &= 6 \cos\left(\frac{5t}{3} + \alpha\right) \\ \text{when } t=0, x &= 6 \\ 6 &= 6 \cos \alpha \\ \cos \alpha &= 1 \\ \alpha &= 0 \\ \therefore x &= 6 \cos \frac{5t}{3} \rightarrow \textcircled{1} \end{aligned}$$

$$\dot{x} = -10 \sin \frac{5t}{3}$$

$$\ddot{x} = -\frac{50}{3} \cos \frac{5t}{3}$$

$$= -\frac{50}{3} \cos 5$$

$$= -4.7277 \dots \rightarrow \textcircled{1}$$

$$= 4.73 \text{ cm/s}^2 \text{ in}$$

the negative direction

(Third mark for correct statement) $\rightarrow \textcircled{1}$

b)

$$\begin{aligned} \ddot{x} &= 0 \\ \dot{x} &= c \end{aligned}$$

$$\begin{aligned} \text{when } x=0, \dot{x} &= 12 \cos(-\theta) \rightarrow \\ &= 12 \cos \theta. \end{aligned}$$

$$\therefore c = 12 \cos \theta.$$

$$\dot{x} = 12 \cos \theta \quad \text{wrong}$$

$$x = 12 \cos \theta + c$$

$$\text{when } t=0, x=0 \therefore c=0$$

$$\therefore x = 12 \cos \theta \rightarrow \textcircled{1}$$

$$\begin{aligned} \ddot{y} &= -10 \\ \dot{y} &= -10t + c \end{aligned}$$

$$\begin{aligned} \text{when } t=0, \dot{y} &= 12 \sin(-\theta) \\ &= -12 \sin \theta \end{aligned}$$

$$\therefore c = -12 \sin \theta.$$

$$\dot{y} = -10t - 12 \sin \theta \quad \text{wrong}$$

$$y = -5t^2 - 12 \sin \theta + c$$

$$\text{when } t=0, y=0 \therefore c=0$$

$$\therefore y = -12 \sin \theta - 5t^2 \rightarrow \textcircled{1}$$

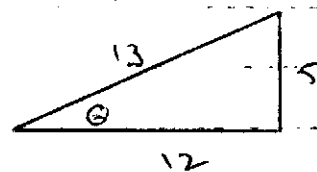
$$\text{(iv)} \quad -3 = -12 \sin \theta - 5t^2$$

$$5t^2 + 12 \sin \theta - 3 = 0$$

$$\text{wrong}$$

$$\sin \theta = \frac{5}{13}$$

$$\cos \theta = \frac{12}{13}$$



$$5t^2 + \frac{60t}{13} - 3 = 0$$

$$65t^2 + 60t - 39 = 0 \rightarrow \textcircled{1}$$

Qn. 5 (cont.)

$$t = \frac{-60 \pm \sqrt{60^2 - 4 \cdot 65 \cdot (-39)}}{130}$$

$$t = \frac{-60 \pm 117.2177461}{130}$$

$$t = 0.4401 \text{ s. or } \left(\begin{array}{l} \text{negative} \\ \text{answer} \\ \text{which is} \\ \text{not required} \end{array} \right) \rightarrow \textcircled{1}$$

$$x = 12t \cos \theta$$

$$= 12 \times 0.4401 \times \frac{12}{13}$$

$$= 4.875 \text{ m} \rightarrow \textcircled{1}$$

(3 dp)

QUESTION 6

(a) (i) $m = p$

$$y - ap^2 = p(x - 2ap)$$

when $x = 0$, $y = -ap^2$

$$Q(0, -ap^2) \rightarrow \textcircled{1}$$

$$\begin{aligned} \text{(iii)} \quad SP^2 &= (2ap)^2 + (ap^2 - a)^2 \\ &= 4a^2p^2 + a^2(p^2 - 1)^2 \\ &= a^2(p^2 + p^4 - 2p^2 + 1) \\ &= a^2(p^4 + 2p^2 + 1) \\ &= a^2(p^2 + 1)^2 \end{aligned}$$

$$\therefore SP = a(p^2 + 1) \quad \left. \begin{aligned} &SQ = a + ap^2 \\ &\therefore SP = SQ \\ &\therefore \Delta_{SPQ} \text{ isosceles} \\ &\therefore \angle SPQ = \angle SQP \\ &(\text{equal angles of } \triangle SPQ) \end{aligned} \right\} \rightarrow \textcircled{1}$$

$$SQ = a + ap^2$$

$$\therefore SP = SQ$$

$$\therefore \Delta_{SPQ} \text{ isosceles}$$

$$\therefore \angle SPQ = \angle SQP$$

$$(\text{equal angles of } \triangle SPQ) \rightarrow \textcircled{1}$$

(b) $\frac{dy}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -e^{-x}$

$$\therefore \frac{1}{2} v^2 = \int -e^{-x} dx$$

$$\frac{1}{2} v^2 = e^{-x} + c \rightarrow \textcircled{1}$$

when $x = 0$, $v = \sqrt{2}$

$$1 = 1 + c$$

$$c = 0$$

$$\therefore \frac{1}{2} v^2 = e^{-x}$$

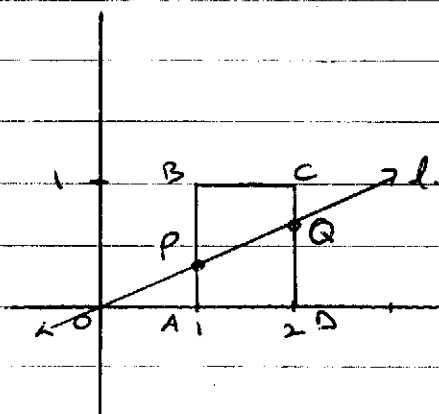
$$v^2 = 2e^{-x}$$

$$v = \pm \sqrt{2} e^{-x/2}$$

but $v = \sqrt{2}$ when $x = 0$

$$\therefore v = \sqrt{2} e^{-x/2} \rightarrow \textcircled{1}$$

(c)



(i) $m_{OD} = 0$

$$m_{OC} = \frac{1}{2}$$

$$\therefore 0 \leq m \leq \frac{1}{2} \rightarrow \textcircled{1}$$

(ii) $P(1, m)$ $Q(2, 2m)$

$$\text{Area of trapezium } APQD = \frac{3m}{2}$$

$$\text{Area of trapezium } PBCQ$$

$$= 1 - \frac{3m}{2}$$

$\rightarrow \textcircled{1}$

$$\therefore \frac{\frac{3m}{2}}{1 - \frac{3m}{2}} = \frac{2}{1}$$

$$\frac{3m}{2} = 2 - 3m$$

$$\frac{9m}{2} = 2$$

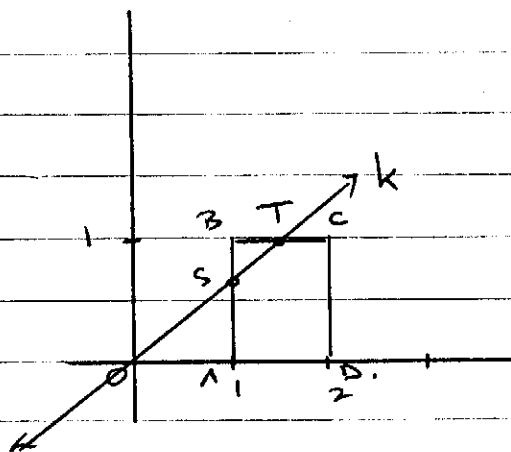
$$m = \frac{4}{9} \rightarrow \textcircled{1}$$

or $\frac{\frac{3m}{2}}{1 - \frac{3m}{2}} = \frac{1}{2}$

$$\therefore m = \frac{2}{9} \rightarrow \textcircled{1}$$

QUESTION 6 (cont.)

(11)



$$S(1, n) \quad T\left(\frac{1}{n}, 1\right)$$

$$\text{Area of } \triangle_{SBT} = \frac{1}{2}(1-n)\left(\frac{1}{n}-1\right)$$

$$\text{Area of } ASTCD = 1 - \frac{1}{2}(1-n)\left(\frac{1}{n}-1\right)$$

$$\text{Prove that } \frac{\text{Area of } \triangle_{SBT}}{\text{Area of } ASTCD} \neq \frac{1}{2}$$

$$\text{LHS} = \frac{\frac{1}{2}(1-n)\left(\frac{1}{n}-1\right)}{1 - \frac{1}{2}(1-n)\left(\frac{1}{n}-1\right)} \rightarrow \textcircled{1}$$

$$= \frac{(1-n)\left(\frac{1}{n}-1\right)}{2 - (1-n)\left(\frac{1}{n}-1\right)}$$

$$= \frac{\frac{1}{n} - 1 - 1 + n}{2 - \left(\frac{1}{n} - 1 - 1 + n\right)}$$

$$= \frac{1 - 2n + n^2}{4n - 1 - n^2}$$

$$\text{Let } \frac{1 - 2n + n^2}{4n - 1 - n^2} = \frac{1}{2} \rightarrow \textcircled{1}$$

$$2 - 4n + 2n^2 = 4n - 1 - n^2$$

$$3n^2 - 8n + 3 = 0$$

$$n = \frac{8 \pm \sqrt{64 - 4 \cdot 3 \cdot 3}}{6}$$

$$= 2.215 \dots$$

or

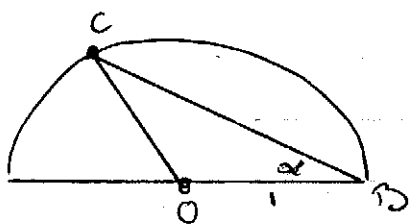
$$0.4514 \dots$$

$$\text{but } \frac{1}{2} \leq n \leq 1 \rightarrow \textcircled{1}$$

$\therefore$ it is not possible for k to divide the area of the square in the ratio 2:1

QUESTION 7

(a)



$$(i) \angle BOC = \pi - 2\alpha \text{ (or } 180 - 2\alpha) \rightarrow \textcircled{1}$$

$$\begin{aligned} (ii) A &= \frac{1}{2} r^2 (\theta - \sin \theta) \\ &= \frac{1}{2} (\pi - 2\alpha - \sin(\pi - 2\alpha)) \\ &= \frac{1}{2} (\pi - 2\alpha - \sin 2\alpha) \end{aligned} \rightarrow \textcircled{1}$$

$$\begin{aligned} (iii) \text{Area of segment} &= \frac{1}{2} \times \text{Area of semi-circle} \\ \frac{1}{2} (\pi - 2\alpha - \sin 2\alpha) &= \frac{1}{4} \pi \rightarrow \textcircled{1} \end{aligned}$$

$$2(\pi - 2\alpha - \sin 2\alpha) = \pi$$

$$2\pi - 4\alpha - 2\sin 2\alpha - \pi = 0$$

$$2\sin 2\alpha + 4\alpha - \pi = 0 \rightarrow \textcircled{1}$$

$$\begin{aligned} (iv) f(x) &= 2\sin 2x + 4x - \pi \\ f(0.4) &= 2\sin 0.8 + 4(0.4) - \pi \\ &= -0.107 < 0 \\ f(0.5) &= 2\sin 1 + 2 - \pi \\ &= 0.541 > 0 \end{aligned} \rightarrow \textcircled{1}$$

Root lies between 0.4 and 0.5
as there is a change of sign.
 $\rightarrow \textcircled{1}$

$$\begin{aligned} (v) f(0.45) &= 2\sin 0.9 + 4(0.45) - \pi \\ &= 0.2251 > 0 \end{aligned}$$

$\therefore$ Root lies closer to 0.4
than 0.5
 $\rightarrow \textcircled{1}$

$$\begin{aligned} (b) (i) \text{ In } \triangle ACD \text{ and } \triangle ADS, \\ \angle ADC &= \angle ASD = 90^\circ \text{ (given)} \\ \angle DAC &= \angle SAD \text{ (common)} \\ \therefore \angle ACD &= \angle ADS = \theta \\ \text{(angle sum of } \triangle) &\rightarrow \textcircled{1} \end{aligned}$$

$$\begin{aligned} (ii) \text{ In } \triangle ACD, \sin \theta &= \frac{AD}{AC} = \frac{AD}{d} \\ \therefore AD &= d \sin \theta \rightarrow \end{aligned}$$

$$\text{In } \triangle ADS, \cos \theta = \frac{DS}{AD}$$

$$\begin{aligned} DS &= AD \cos \theta \\ &= d \sin \theta \cos \theta \end{aligned}$$

$$\therefore \text{Distance} = d \sin \theta (1 + \cos \theta) \rightarrow \textcircled{1}$$

$$\begin{aligned} (iii) \angle DSF &= \theta \\ \cos \theta &= \frac{SF}{DS} \\ SF &= DS \cos \theta \\ &= d \sin \theta \cos^2 \theta \end{aligned}$$

$\therefore$ Geometric series with
 $a = d \sin \theta$ and $r = \cos \theta$
 $\rightarrow \textcircled{1}$

Limiting sum as $0 < |r| < 1$

$$\begin{aligned} S &= \frac{a}{1-r} \\ &= \frac{d \sin \theta}{1 - \cos \theta} \end{aligned} \rightarrow \textcircled{1}$$